

AWS Certified Cloud Practitioner

One-Day Exam Preparation Guide

Exam Code: **CLF-C02** | 65 Questions | 90 Minutes | Pass Score: 700/1000

Domain 1: Cloud Concepts **24%** Domain 2: Security & Compliance **30%** Domain 3: Cloud Technology **34%** Domain 4: Billing & Pricing **12%**

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1 Domain 1 — Cloud Concepts (24%)

1.1 What is Cloud Computing?

Cloud computing is the **on-demand delivery of IT resources over the internet with pay-as-you-go pricing**. Instead of buying and maintaining physical hardware, you access compute power, storage, and databases when you need them.

Six Advantages of Cloud Computing

1. **Trade CapEx for OpEx** — Pay only for what you consume; no upfront hardware investment.
2. **Massive economies of scale** — AWS's aggregate usage lowers per-unit cost for everyone.
3. **Stop guessing capacity** — Scale up or down in minutes; eliminate over/under provisioning.
4. **Increase speed & agility** — New IT resources available in minutes, not weeks.
5. **Stop spending on data-center operations** — Focus on projects that differentiate your business.
6. **Go global in minutes** — Deploy to multiple AWS Regions with a few clicks.

1.2 Service Models

Model	You manage	AWS manages	Examples
IaaS	OS, runtime, middleware, data, apps	Physical servers, network, storage	EC2, VPC, EBS
PaaS	Data & applications	Everything below app layer	Elastic Beanstalk, RDS
SaaS	Nothing (just use the app)	Everything	Gmail, Salesforce, Zoom

1.3 Deployment Models

Public Cloud All resources on AWS infrastructure; fully managed by AWS.

Private Cloud Cloud infrastructure operated solely for one organisation; on-premises.

Hybrid Cloud Mix of on-premises (or private cloud) + public cloud, connected via VPN or Direct Connect.

Multi-Cloud Using services from more than one cloud provider simultaneously.

1.4 AWS Global Infrastructure

Region → Availability Zone (AZ) → Data Center → Edge Location

Region A geographical area with **2 or more AZs** (e.g., `us-east-1`). Choose based on: compliance, latency, service availability, pricing.

Availability Zone (AZ) One or more discrete data centers with redundant power, networking, and connectivity. AZs within a Region are connected by low-latency links. Deploy across ≥ 2 AZs for High Availability.

Edge Location / PoP Used by **CloudFront (CDN)** and **Route 53** to cache content closer to end users. More numerous than Regions.

Local Zone Extension of an AWS Region placed in a metropolitan area for ultra-low latency.

Wavelength Zone Embeds AWS compute inside telecom networks for 5G edge workloads.

AWS Outposts Fully managed AWS infrastructure delivered on-premises.

Exam tip: Region vs AZ: a Region is a city/country; an AZ is a building within that city.

1.5 Cloud Economics: CapEx vs. OpEx

	CapEx (Traditional)	OpEx (Cloud)
Definition	Large upfront investment	Pay-as-you-go operational cost
Tax benefit	Depreciated over time	Deducted in the same year
Flexibility	Fixed capacity	Elastic, adjust anytime
Risk	High (over/under provision)	Low (right-size as needed)

1.6 Well-Architected Framework — 6 Pillars

Mnemonic: **OSRPCS** — Operational Excellence, Security, Reliability, Performance Efficiency, Cost Optimization, Sustainability

Operational Excellence Run and monitor systems to deliver business value; automate with **CloudFormation**, **CodePipeline**.

Security Protect data and systems; use **IAM**, **KMS**, **CloudTrail**, **GuardDuty**.

Reliability Recover from disruptions; design for fault tolerance with Multi-AZ, Auto Scaling, Route 53 health checks.

Performance Efficiency Use computing resources efficiently; choose right instance types, use **ElastiCache**, **CloudFront**.

Cost Optimization Eliminate waste; use Reserved Instances, Spot, **Cost Explorer**, right-sizing.

Sustainability Minimize environmental impact; use managed services, right-size, select efficient regions.

1.7 Cloud Architecture Design Principles

- **Design for failure** — Assume things will fail; build redundancy.
- **Decouple components** — Use SQS, SNS to separate tightly coupled services.
- **Implement elasticity** — Auto Scaling, serverless.
- **Think parallel** — Use horizontal scaling, not vertical.
- **Stop guessing capacity** — Provision dynamically.

2 Domain 2 — Security & Compliance (30%)

2.1 Shared Responsibility Model

AWS is responsible FOR the cloud. YOU are responsible IN the cloud.

Layer	AWS Responsibility	Customer Responsibility
Physical	Data centers, hardware	Nothing
Network infrastructure	Global backbone	VPC config, security groups
Hypervisor	Managed by AWS	Nothing
OS (EC2)	None	Patch & harden guest OS
OS (RDS/Lambda)	AWS patches it	Nothing
Applications	None	Code, config, updates
Data	Encryption tools provided	Classify, encrypt, manage data
IAM	AWS IAM service	Users, roles, policies you create

Exam tip: “Security OF the cloud” = AWS. “Security IN the cloud” = you. Services like Lambda/RDS shift more OS responsibility to AWS.

2.2 Identity & Access Management (IAM)

IAM is **global** (not region-specific) and controls *who* can do *what* to *which* AWS resources.

Core IAM Concepts

Root Account The email used to create the AWS account. **Never use for daily tasks. Enable MFA immediately.**

IAM User A person or application with long-term credentials (username/password or access keys).

IAM Group A collection of users; apply policies to the group, not individual users.

IAM Role Temporary credentials; assumed by users, services, or federated identities. **Best practice for EC2 to access S3, etc.**

IAM Policy JSON document that defines Allow/Deny for Actions on Resources under Conditions.

Policy Evaluation Logic

1. Default: **implicit DENY** on everything.
2. Explicit **DENY** overrides any Allow.
3. Explicit **ALLOW** grants access (unless there's also an explicit Deny).

IAM Best Practices

- Enable MFA on root and privileged accounts.
- Grant **least privilege** — only permissions needed for the task.
- Use roles instead of embedding access keys.
- Rotate credentials regularly; use **IAM Credential Report** to audit.
- Never share access keys; use IAM roles for EC2 instances.

IAM Policy Types

Type	Description
Identity-based	Attached to user, group, or role
Resource-based	Attached to resource (S3 bucket policy, SNS topic policy)
Service Control Policy (SCP)	Applied via AWS Organizations; limits max permissions for member accounts
Permission Boundary	Max permissions an identity can have; overrides identity-based policies
Session Policy	Passed when assuming a role via API

2.3 AWS Security Services

Service	Purpose & Key Facts
IAM	Authentication & authorization; global service
AWS Organizations	Manage multiple accounts; SCPs; consolidated billing
AWS Control Tower	Best-practice multi-account setup with guardrails
AWS SSO / IAM Identity Center	Single sign-on for multiple accounts and applications
Amazon Cognito	User identity for web/mobile apps; user pools + identity pools
AWS KMS	Create and manage encryption keys; integrated with most AWS services
AWS CloudHSM	Hardware Security Module in your VPC; FIPS 140-2 Level 3; you manage keys
AWS Secrets Manager	Store, rotate, retrieve database passwords and API keys automatically
AWS Certificate Manager (ACM)	Provision and manage SSL/TLS certificates for free
AWS Shield Standard	Free DDoS protection on all AWS accounts automatically
AWS Shield Advanced	Paid; 24/7 DDoS response team, financial protection, detailed attack metrics
AWS WAF	Web Application Firewall; filter HTTP(S) based on rules; protects CloudFront, ALB, API GW
Amazon GuardDuty	Threat detection using ML; analyzes CloudTrail, VPC Flow Logs, DNS logs
Amazon Inspector	Automated vulnerability assessment for EC2 and container images
Amazon Macie	Uses ML to discover and protect sensitive data (PII) in S3
AWS Security Hub	Centralised security alerts and compliance findings aggregator
Amazon Detective	Investigate security incidents; graph analysis of logs
AWS Firewall Manager	Centrally manage WAF rules, Shield Advanced, security groups across accounts
AWS Network Firewall	Managed stateful network firewall for VPC traffic filtering

2.4 Logging & Auditing

AWS CloudTrail Logs **every API call** made in your AWS account (who did what, when, from where). Enabled by default for 90 days; create a Trail to S3 for long-term retention. **Governance, compliance, and auditing.**

Amazon CloudWatch Metrics, logs, alarms, dashboards. Monitors performance and operational health.

AWS Config Records **configuration changes** to resources over time; evaluate against compliance rules.

VPC Flow Logs Capture IP traffic metadata (accept/reject) for VPC, subnet, or ENI.

AWS CloudTrail Insights Automatically detects unusual API activity.

Exam tip: CloudTrail = WHO did WHAT. CloudWatch = WHAT is happening now (metrics). Config = HOW did it CHANGE.

2.5 Data Protection & Encryption

Encryption at rest KMS-managed keys (SSE-KMS), S3-managed keys (SSE-S3), customer keys (SSE-C).

Encryption in transit TLS/SSL; use ACM to provision certificates.

S3 Object Lock WORM (Write-Once-Read-Many); Compliance or Governance mode.

S3 Versioning Keep multiple versions of an object; protects against accidental deletion.

2.6 Compliance Programs

AWS maintains compliance with **PCI-DSS, HIPAA, SOC 1/2/3, ISO 27001, GDPR, FedRAMP** and many others.

AWS Artifact Self-service portal to download compliance reports (SOC, PCI) and agreements (BAA, NDA) — no cost.

AWS Compliance Center Country-specific cloud compliance information.

3 Domain 3 — Cloud Technology & Services (34%)

3.1 Compute Services

Amazon EC2 (Elastic Compute Cloud)

Virtual machines in the cloud. You choose OS, instance type, storage, and networking.

Instance Families:

Family	Type	Use case
t3/t4g	General purpose	Dev, test, low-traffic apps
m6i/m7g	General purpose	Web servers, databases
c6i/c7g	Compute optimised	HPC, ML inference, batch
r6i/r7g	Memory optimised	In-memory DBs, real-time analytics
x2idn	Memory optimised	SAP HANA, large in-memory
p4/g5	Accelerated (GPU)	ML training, video rendering
i3/i4i	Storage optimised	NoSQL, data warehouses

EC2 Purchasing Options:

Option	Discount	When to use
On-Demand	Baseline	Short-term, unpredictable, testing
Reserved Instance (RI)	Up to 72%	Predictable, 1 or 3 year commitment
Savings Plans	Up to 72%	Like RI but more flexible (Compute/EC2/SageMaker)
Spot Instance	Up to 90%	Fault-tolerant, flexible, batch jobs
Dedicated Host	High	Compliance, BYOL (Bring Your Own License)
Dedicated Instance	Medium	Isolated hardware, avoid multi-tenancy

Key EC2 concepts: AMI (Amazon Machine Image) = OS template. Security Group = stateful virtual firewall at instance level. Key Pair = SSH access. User Data = bootstrap script run at launch. Instance Metadata = 169.254.169.254 (info about running instance).

Auto Scaling & Load Balancing

Auto Scaling Group (ASG) Maintains desired number of EC2 instances; scales out/in based on CloudWatch alarms. Define: min, desired, max capacity.

Application Load Balancer (ALB) Layer 7 (HTTP/HTTPS); route by path, host, headers; supports WebSocket; ideal for microservices.

Network Load Balancer (NLB) Layer 4 (TCP/UDP); ultra-low latency; handles millions of requests per second; supports static IP / Elastic IP.

Gateway Load Balancer (GWLb) Distributes traffic to virtual appliances (firewalls, IDS/IPS) at Layer 3.

Classic Load Balancer (CLB) Legacy; not recommended for new applications.

Serverless Compute

AWS Lambda Run code without provisioning servers. Pay per invocation & duration (ms). Max 15 min timeout. Integrates with 200+ services. Supports Node.js, Python, Java, Go, Ruby, .NET.

AWS Fargate Serverless compute engine for **containers** (works with ECS and EKS). No EC2 management.

AWS App Runner Fully managed service to build and run containerised web apps from source code or container image.

Container Services

Amazon ECS Elastic Container Service; AWS-native container orchestration.

Amazon EKS Elastic Kubernetes Service; managed Kubernetes.

Amazon ECR Elastic Container Registry; Docker image repository.

Other Compute

AWS Elastic Beanstalk PaaS — upload code (Java, .NET, PHP, Node, Python, Ruby, Go, Docker); AWS handles deployment, capacity, LB, auto scaling. You still have access to underlying EC2.

AWS Lightsail Simple VPS for small apps, blogs, websites. Fixed monthly price. Includes compute, storage, DNS, IP.

AWS Batch Fully managed batch computing; runs jobs on EC2 or Fargate; schedules and provisions automatically.

AWS Outposts AWS-managed hardware in your own data center; run AWS services on-premises.

3.2 Storage Services

Amazon S3 (Simple Storage Service)

S3 is **object storage** (flat namespace, not a file system). Unlimited storage. Objects up to **5 TB**. Buckets are Region-specific but names are globally unique.

S3 Storage Classes:

Class	Avail.	Use Case
S3 Standard	99.99%	Frequently accessed; default choice
S3 Intelligent-Tiering	99.9%	Unknown/changing access; auto-moves between tiers
S3 Standard-IA	99.9%	Infrequent access; retrieval fee applies
S3 One Zone-IA	99.5%	Non-critical data; lower cost, single AZ
S3 Glacier Instant Retrieval	99.9%	Archive; millisecond retrieval
S3 Glacier Flexible Retrieval	99.99%	Archive; minutes to hours retrieval
S3 Glacier Deep Archive	99.99%	Long-term archive; 12–48 hr retrieval; cheapest

S3 Key Features:

- **Versioning:** Keep all versions of objects; protects against accidental deletes.
- **Lifecycle Policies:** Automatically transition objects between storage classes.
- **Replication:** CRR (Cross-Region Replication) for DR; SRR (Same-Region) for compliance.
- **S3 Transfer Acceleration:** Upload via CloudFront edge locations.
- **Multipart Upload:** Upload large objects in parts (recommended >100 MB, required >5 GB).
- **Pre-signed URLs:** Time-limited URLs to grant temporary access to private objects.
- **Static Website Hosting:** Host static HTML/CSS/JS websites directly from S3.
- **Object Lock:** WORM protection; Compliance or Governance mode.
- **Access Points:** Simplified access management for shared datasets.

Block & File Storage

Amazon EBS (Elastic Block Store) Persistent block storage for EC2. Tied to one AZ. Like a hard drive. Snapshot to S3 for backup. Types: gp3 (general), io2 (high IOPS), st1 (throughput), sc1 (cold).

Amazon EFS (Elastic File System) Managed NFS file system shared across multiple EC2 instances and AZs. Scales automatically. Linux only.

Amazon FSx for Windows Managed Windows file server (SMB/NTFS/AD integration).

Amazon FSx for Lustre High-performance file system for HPC and ML workloads; integrates with S3.

Data Transfer & Hybrid Storage

AWS Storage Gateway Hybrid cloud storage bridge: File, Volume, or Tape Gateway types. On-premises apps connect to AWS storage.

AWS DataSync Automated online data transfer between on-premises storage/other clouds and AWS. Uses TLS encryption.

AWS Snow Family Physical devices to transfer large amounts of data offline.

Device	Capacity & Use
Snowcone	8 TB HDD / 14 TB SSD; smallest; edge computing
Snowball Edge (Storage)	80 TB; petabyte-scale migration
Snowball Edge (Compute)	42 TB storage + compute power; edge ML
Snowmobile	Exabyte scale; actual truck; 100 PB per trip

3.3 Database Services

Service	Type	Best For
RDS	Relational (SQL)	MySQL, PostgreSQL, MariaDB, Oracle, SQL Server; managed; Multi-AZ
Aurora	Relational (SQL)	MySQL/PostgreSQL-compatible; 5x faster; 6 copies across 3 AZs; Serverless option
DynamoDB	NoSQL (key-value)	Single-digit millisecond latency; serverless; unlimited scale
Redshift	Data warehouse	Columnar; analytics/BI; petabyte scale; use Redshift Spectrum for S3 queries
ElastiCache	In-memory cache	Redis or Memcached; sub-millisecond latency; relieve DB load
Amazon Neptune	Graph database	Social networks, knowledge graphs, fraud detection
Amazon QLDB	Ledger database	Immutable, cryptographically verifiable transaction log
Amazon DocumentDB	Document (NoSQL)	MongoDB-compatible; JSON data
Amazon Keyspaces	Wide-column (NoSQL)	Apache Cassandra-compatible
Amazon Timestream	Time-series	IoT, DevOps metrics

RDS Multi-AZ Synchronous standby in a second AZ; automatic failover; for **high availability**, not read scaling.

RDS Read Replicas Asynchronous copy; for **read scaling**; can be cross-region.

DynamoDB DAX In-memory cache for DynamoDB; microsecond reads.

AWS DMS Database Migration Service; migrate on-prem or cloud DBs to AWS; homogeneous & heterogeneous.

AWS Schema Conversion Tool Convert schema from one DB engine to another (e.g., Oracle → Aurora).

3.4 Networking Services**Amazon VPC (Virtual Private Cloud)**

VPC = your own isolated network in AWS. You control IP ranges (CIDR), subnets, route tables, internet gateways, NAT, security groups, and NACLs.

Subnet A range of IPs within a VPC. **Public subnet**: has route to Internet Gateway. **Private subnet**: no direct internet route.

Internet Gateway (IGW) Allows resources in public subnets to access the internet (two-way).

NAT Gateway Allows **private subnet** resources to initiate outbound internet traffic; not inbound. Managed by AWS. Place in public subnet.

Security Group Stateful firewall at the **instance/ENI** level. Allow rules only; return traffic auto-allowed.

Network ACL (NACL) Stateless firewall at the **subnet** level. Allow and Deny rules; both inbound and outbound must be defined.

VPC Peering Connect two VPCs privately; non-transitive (A–B–C: A cannot reach C via B).

AWS Transit Gateway Hub-and-spoke to connect thousands of VPCs and on-premises networks. Transitive routing.

VPC Endpoints Private connection from VPC to AWS services without internet. **Interface endpoint** (most services, uses ENI). **Gateway endpoint** (S3 and DynamoDB only, free).

VPC Flow Logs Capture network traffic metadata for analysis.

Elastic IP Static public IPv4 address; assigned to your account until released.

DNS & Content Delivery

Amazon Route 53 Managed DNS service. Routing policies: Simple, Weighted, Latency-based, Failover, Geolocation, Geoproximity, Multi-value. Also does domain registration.

Amazon CloudFront Global Content Delivery Network (CDN). Caches content at 400+ edge locations. Reduces latency, absorbs DDoS, integrates with Shield & WAF. Supports HTTPS, custom domains.

Hybrid Connectivity

AWS VPN (Site-to-Site) Encrypted tunnel over internet between on-premises and VPC. Quick to set up; variable latency.

AWS Direct Connect Dedicated private physical connection (1/10/100 Gbps) from on-premises to AWS. Consistent latency, no internet traversal; takes weeks to provision.

AWS Direct Connect + VPN Direct Connect for bandwidth, VPN for encryption — highly secure hybrid connectivity.

3.5 Management & Monitoring

Monitoring

Amazon CloudWatch Collect metrics, create alarms, dashboards, log groups. **CloudWatch Logs Insights**: query logs. **CloudWatch Events/EventBridge**: react to events (e.g., EC2 state change triggers Lambda).

AWS CloudTrail Log all AWS API calls; store to S3; detect suspicious activity.

AWS Config Track resource configurations over time; evaluate against rules; view compliance history.

AWS Trusted Advisor Checks across 5 categories: Cost Optimization, Performance, Security, Fault Tolerance, Service Limits. Free tier: 7 core checks. Business/Enterprise: all checks.

AWS Health Dashboard Service Health (public) and Account Health (personal) notifications.

Infrastructure as Code & Automation

AWS CloudFormation IaC using JSON/YAML **templates**. Deploys a **stack** of resources. Repeatable, version-controlled. No extra charge (pay for resources created).

AWS CDK Cloud Development Kit; define infrastructure in TypeScript, Python, Java — compiles to CloudFormation.

AWS Systems Manager Manage EC2 at scale: **Session Manager** (no SSH needed), **Parameter Store** (config/secrets), **Patch Manager**, Run Command, Automation.

AWS OpsWorks Managed Chef and Puppet for configuration management.

Developer Tools

AWS CodeCommit Managed Git repository (like GitHub on AWS). *Note: being retired June 2024.*

AWS CodeBuild Fully managed CI build service; compiles code, runs tests, produces artifacts.

AWS CodeDeploy Automated deployments to EC2, Lambda, ECS, on-premises.

AWS CodePipeline Fully managed CI/CD pipeline orchestrator.

AWS CodeStar (Legacy) Unified UI for CI/CD project management.

AWS X-Ray Distributed tracing; analyse and debug microservices and serverless apps.

Amazon CodeGuru ML-powered code review and performance profiling.

3.6 Application Integration

Amazon SQS Simple Queue Service; **decouples** services; Standard (best-effort ordering) vs FIFO (exactly-once, ordered, 3000 msg/s). Messages retained up to 14 days.

Amazon SNS Simple Notification Service; **pub/sub model**; push to many subscribers (email, SQS, Lambda, HTTP, SMS). Fan-out pattern: SNS → multiple SQS.

Amazon EventBridge Serverless event bus; route events from AWS services, SaaS apps, custom apps to targets. Successor to CloudWatch Events.

AWS Step Functions Orchestrate multi-step workflows as state machines; coordinate Lambda, ECS, DynamoDB.

Amazon MQ Managed ActiveMQ/RabbitMQ; for apps using JMS/AMQP/STOMP protocols.

Amazon Kinesis Real-time streaming data. **Kinesis Data Streams**: ingest and store. **Kinesis Firehose**: load to S3/Redshift/OpenSearch. **Kinesis Analytics**: SQL on streams.

Amazon API Gateway Create, publish, and manage REST, HTTP, and WebSocket APIs. Integrates with Lambda for fully serverless APIs.

AWS AppSync Managed GraphQL service.

3.7 Analytics Services

Service	Purpose
Amazon Athena	Serverless SQL queries on S3 data; pay per query; uses Presto engine
Amazon Redshift	Cloud data warehouse; columnar; BI and analytics
Amazon EMR	Managed Hadoop/Spark/Hive cluster for big data processing
AWS Glue	Serverless ETL; data catalog; crawler discovers schema in S3/RDS
Amazon QuickSight	Business Intelligence (BI) and data visualization tool; ML insights
Amazon OpenSearch	Managed Elasticsearch; log analytics, search
AWS Data Exchange	Subscribe to third-party data sets
AWS Lake Formation	Build secure data lakes on S3

3.8 Machine Learning & AI Services

Service	Capability
Amazon SageMaker	Build, train, and deploy ML models at scale
Amazon Rekognition	Image and video analysis (object detection, facial recognition)
Amazon Transcribe	Speech-to-text
Amazon Translate	Neural machine translation
Amazon Polly	Text-to-speech
Amazon Comprehend	NLP: sentiment analysis, entity detection, key phrases
Amazon Lex	Build conversational chatbots (powers Alexa)
Amazon Kendra	ML-powered intelligent search engine
Amazon Textract	OCR + extract structured data from documents/forms
Amazon Forecast	Time-series forecasting with ML
Amazon Fraud Detector	Detect online fraud with ML
Amazon Personalize	Real-time personalisation and recommendation engine
Amazon Bedrock	Foundation models (FMs) as a service; build generative AI apps
Amazon Q	Generative AI assistant for business; integrates with your data

3.9 Migration & Transfer

The 7 Rs of Migration

1. **Retire** — Decommission unused apps.

2. **Retain** — Keep on-premises (not ready to migrate).
3. **Rehost** (Lift & Shift) — Move as-is to EC2; minimal change. Fastest migration.
4. **Relocate** — Move VMware workloads to VMware Cloud on AWS.
5. **Repurchase** — Move to a different product (e.g., CRM → Salesforce).
6. **Replatform** (Lift, Tinker & Shift) — Minor optimizations during migration (e.g., migrate DB to RDS).
7. **Refactor / Re-architect** — Re-design using cloud-native; highest effort, most benefit.

Migration Services

AWS Application Migration Service (MGN) Lift-and-shift migration; replicate servers to AWS continuously, then cutover.

AWS Migration Hub Central dashboard to track migration progress.

AWS Application Discovery Service Discover on-premises servers, dependencies, and performance for migration planning.

AWS DMS Migrate databases; homogeneous (MySQL → RDS MySQL) or heterogeneous (Oracle → Aurora).

AWS SCT Schema Conversion Tool; converts DB schemas between engines.

3.10 End-User Computing & Other

Amazon WorkSpaces Managed cloud desktops (Virtual Desktop Infrastructure).

Amazon AppStream 2.0 Stream desktop applications to a browser; no installation needed.

Amazon WorkDocs Managed document collaboration.

AWS IoT Core Connect and manage billions of IoT devices; MQTT protocol.

AWS Ground Station Managed satellite communication.

Amazon Connect Cloud contact center (call center) solution.

Amazon Chime Communication service (meetings, chat, calls).

4 Domain 4 — Billing, Pricing & Support (12%)

4.1 AWS Pricing Fundamentals

Three fundamentals of AWS pricing:

1. **Compute** — Pay for time resources are running (per-second or per-hour billing).
2. **Storage** — Pay for the amount of data stored (per GB/month).
3. **Data transfer** — Inbound to AWS: **FREE**. Outbound to internet: charged. Transfer between AZs: charged. Within same AZ: free.

Free Tier: Three types of free offers:

- **Always free:** Lambda (1M req/month), DynamoDB (25 GB), CloudWatch (10 metrics).
- **12 months free:** EC2 (750 hrs/month t2.micro), S3 (5 GB), RDS (750 hrs db.t2.micro).
- **Trials:** Short-term free trials for specific services.

4.2 Cost Management Tools

AWS Pricing Calculator Estimate monthly costs before deploying; build architecture scenarios.

AWS Cost Explorer Visualise, understand, and manage AWS costs and usage over time; 12 months historical. Forecasting capability. Identify Reserved Instance recommendations.

AWS Budgets Set custom cost/usage/RI/Savings Plan budgets and receive **alerts** when thresholds are exceeded.

AWS Cost and Usage Report (CUR) Most comprehensive billing data; CSV to S3; use with Athena/QuickSight for analysis.

AWS Cost Allocation Tags Tag resources (e.g., **Project=Alpha**); track costs by tag in billing reports.

AWS Cost Anomaly Detection ML-powered; automatically detects unexpected spending

patterns.

Savings Plans Flexible pricing model (commit to \$/hour for 1 or 3 years); up to 72% discount. Types: Compute, EC2 Instance, SageMaker.

4.3 Consolidated Billing & AWS Organizations

AWS Organizations Create and manage multiple AWS accounts under a single management account. Features: Consolidated Billing, SCPs, Account Vending (AWS Control Tower).

Consolidated Billing All linked accounts' usage is aggregated; benefits from **volume discounts** and Reserved Instance/Savings Plan sharing across accounts.

Management Account (Formerly master account) Pays for all linked accounts; cannot have SCPs applied to itself.

Service Control Policies (SCPs) Applied at OU or account level; restrict max permissions; do not grant permissions.

Organizational Unit (OU) Group of accounts within an organization for policy management.

Exam tip: Even with consolidated billing, each account's usage appears separately in the CUR. Volume pricing is computed on combined usage.

4.4 AWS Support Plans

Feature	Basic	Developer	Business	Enterprise
Cost	Free	\$29/mo	\$100/mo	\$15K/mo
TAM	No	No	No	Yes
AWS Concierge	No	No	No	Yes
24/7 phone/chat	No	No	Yes	Yes
Business hours email	No	Yes	Yes	Yes
Trusted Advisor	7 checks	7 checks	All	All
Response: critical	–	–	1 hr	15 min
Response: production	–	–	4 hr	1 hr
Response: general	–	24 hr	24 hr	24 hr

TAM (Technical Account Manager) is only in **Enterprise** plan.

24/7 phone support starts at **Business** plan.

Enterprise On-Ramp: \$5.5K/mo; pool of TAMs; 30-min critical response.

4.5 AWS Marketplace

A digital catalog with thousands of software listings from independent software vendors (ISVs). Buy, deploy, and manage third-party software on AWS. Flexible pricing: free, BYOL, hourly, monthly, annual. Sellers include startups, enterprises, and AWS Partners.

4.6 AWS Partner Network (APN)

Consulting Partners Help customers design, build, migrate, and manage workloads on AWS (SI, MSPs).

Technology Partners Software or hardware integrated with AWS (ISVs, SaaS providers).

Tiers Select, Advanced, Premier (based on certifications, revenue, customer success).

5 Key Comparisons & Must-Know Distinctions

5.1 Security Group vs. Network ACL

Feature	Security Group	Network ACL
Level	Instance/ENI	Subnet
Stateful/Stateless	Stateful	Stateless
Rules	Allow only	Allow and Deny
Default (new)	Deny all inbound, allow all outbound	Allow all inbound and outbound
Rule evaluation	All rules evaluated	Rules evaluated in order by number

5.2 CloudWatch vs. CloudTrail vs. Config

	CloudWatch	CloudTrail	AWS Config
What it does	Monitor metrics & logs	Audit API calls	Track config changes
Key question	<i>Is the system healthy?</i>	<i>Who did what, when?</i>	<i>How did this change?</i>
Main use	Ops monitoring, alerting	Compliance, forensics	Compliance, drift detection

5.3 RDS vs. Aurora vs. DynamoDB vs. Redshift

	RDS	Aurora	Dynamo-DB	Redshift
Type	Relational	Relational	NoSQL	Data Warehouse
Engine	6 options	MySQL/PG	Key-value	Columnar
Scaling	Vertical	Auto	Serverless	Node-based
Best for	OLTP	Managed OLTP	Millisecond	Analytics/BI

5.4 S3 vs. EBS vs. EFS

	S3	EBS	EFS
Type	Object	Block	File (NFS)
Access	Via HTTP/SDK	Single EC2 (mostly)	Multi EC2 / multi-AZ
Size limit	Unlimited	Up to 64 TB	Unlimited
Region/AZ	Regional	Single AZ	Multi-AZ
OS	Any	Any	Linux only
Use case	Backup, web, static	Boot volumes, DBs	Shared storage

5.5 SQS vs. SNS vs. EventBridge

	SQS	SNS	EventBridge
Pattern	Queue (pull)	Pub/Sub (push)	Event bus (push)
Consumers	One consumer	Multiple	Multiple targets
Persistence	14 days	No retention	No retention
Use case	Decouple services	Alerts, fan-out	Event-driven arch.

5.6 Lambda vs. EC2 vs. ECS/Fargate

	Lambda	EC2	ECS/Fargate
Server management	None	You manage OS	None (Fargate)
Scaling	Automatic	Manual/ASG	Automatic
Runtime limit	15 minutes	Unlimited	Unlimited
Pricing	Per invocation	Per hour	Per task
Best for	Short, event-driven	Full control needed	Containers

6 Important Numbers & Limits to Memorise

- S3 max object size: **5 TB**
- S3 multipart upload recommended: **>100 MB**
- S3 Glacier Deep Archive retrieval: **12–48 hours**
- Lambda max timeout: **15 minutes**
- Lambda max memory: **10,240 MB (10 GB)**
- Lambda max deployment package: **50 MB (zipped)**
- EBS max size (single volume): **64 TB**
- SQS message retention: **up to 14 days**
- SQS max message size: **256 KB**
- SQS FIFO throughput: **3,000 msg/s with batching**
- DynamoDB item max size: **400 KB**
- CloudTrail event retention (default): **90 days**
- CloudWatch Logs retention: **configurable (1 day – 10 years)**
- Reserved Instance term: **1 or 3 years**
- Spot Instance savings: **up to 90%**
- Reserved Instance savings: **up to 72%**
- Savings Plan savings: **up to 72%**
- Aurora storage copies: **6 copies across 3 AZs**
- RDS automated backup retention: **up to 35 days**
- VPC default CIDR: **/16 max, /28 min**
- SLA for EC2 (Multi-AZ): **99.99%**
- S3 durability: **99.999999999% (11 nines)**
- S3 Standard availability: **99.99%**
- IAM users per account (default): **5,000**

7 Quick Reference Cheat Sheets

7.1 “Which service do I use for...?”

Requirement	Service
Serverless functions	Lambda
Run containers without managing servers	Fargate
PaaS, deploy code without managing infra	Elastic Beanstalk
Host a static website cheaply	S3 Static Website
Global CDN, reduce latency	CloudFront
DNS management and routing	Route 53
Relational managed database	RDS or Aurora
NoSQL, single-digit ms latency	DynamoDB
Analytics / data warehouse	Redshift
Cache to reduce DB load	ElastiCache
Real-time data streaming	Kinesis
Decouple services (queue)	SQS
Push notifications / fan-out	SNS
Event-driven architecture	EventBridge
Dedicated private connection to AWS	Direct Connect
Encrypted tunnel over internet to AWS	Site-to-Site VPN
Connect many VPCs and on-prem	Transit Gateway
Manage secrets / rotate passwords	Secrets Manager
Detect threats / unusual activity	GuardDuty
Find vulnerabilities in EC2	Inspector
Find PII in S3	Macie
Audit API calls	CloudTrail
Monitor metrics and set alarms	CloudWatch
Infrastructure as Code	CloudFormation
Migrate databases	DMS
Move large data offline	Snowball
Automate data transfer online	DataSync
Download compliance reports	Artifact
Visualise costs	Cost Explorer
Set spending alerts	Budgets
Estimate AWS costs before using	Pricing Calculator
Multi-account management	Organizations
ML image recognition	Rekognition
Text-to-speech	Polly
Speech-to-text	Transcribe
Managed Kubernetes	EKS
Managed Docker containers	ECS

7.2 Exam Traps & Common Mistakes

1. **IAM is global** — not tied to any region.
2. **S3 bucket names are globally unique** — but buckets are region-specific.
3. **Security Groups are stateful** — if you allow inbound, outbound reply is auto-allowed.
4. **NACLs are stateless** — you must define both inbound AND outbound rules.
5. **VPC Peering is not transitive** — need a Transit Gateway for hub-and-spoke.
6. **Spot Instances can be interrupted** — not for databases or critical workloads.
7. **RDS Multi-AZ is for HA**, not for read performance — use Read Replicas for reads.

8. **CloudTrail logs API calls**; CloudWatch logs metrics and application logs.
9. **Shield Standard is free** and automatic — Shield Advanced costs extra.
10. **AWS Artifact is free** — it's just a compliance document portal, not a security service.
11. **Trusted Advisor** has full checks only for Business and Enterprise support.
12. **TAM (Technical Account Manager)** is Enterprise support plan only.
13. **NAT Gateway** is in a **public** subnet but serves **private** subnet instances.
14. **Egress-Only Internet Gateway** is NAT for IPv6 (not IPv4).
15. **S3 durability is 11 nines** — availability varies by storage class.
16. **Reserved Instances** can be Standard (sell on marketplace) or Convertible (exchange).
17. **Savings Plans** are more flexible than RIs — commit to \$/hr not specific instance types.
18. **Gateway Endpoints** (S3, DynamoDB) are free; Interface Endpoints cost money.
19. **ACM certificates are free** for use with AWS services (CloudFront, ALB).
20. **Data transfer IN to AWS is always free**; data transfer OUT costs money.

7.3 Shared Responsibility Quick Lookup

Customer Responsibility	AWS Responsibility
IAM users, groups, roles, policies	Physical security of data centers
EC2 operating system patches	Network and hardware infrastructure
Application code and configuration	Hypervisor patching
Security group and NACL rules	Managed service OS (RDS, Lambda)
Data encryption choices	Global network backbone
S3 bucket policies and ACLs	Edge locations
Enabling CloudTrail and Config	Hardware fault management
VPC routing and subnets	AZ redundancy

8 Well-Architected Framework — Deep Dive

Pillar	Key Principle	AWS Tools / Services
Operational Excellence	Automate and iterate; learn from failures	CloudFormation, CodePipeline, CloudWatch, Systems Manager
Security	Apply security at every layer; principle of least privilege	IAM, KMS, CloudTrail, GuardDuty, Shield, WAF
Reliability	Recover from failures; stop guessing capacity; manage change	Multi-AZ, Auto Scaling, Route 53, AWS Backup, RDS Multi-AZ
Performance Efficiency	Use the right resource types; stay updated	CloudFront, ElastiCache, Lambda, right-sizing tools
Cost Optimization	Eliminate waste; right-size; use pricing models	Cost Explorer, Budgets, Reserved Instances, Savings Plans, Spot
Sustainability	Minimize environmental impact; share resources	Serverless, managed services, efficient regions

AWS Well-Architected Tool: Free service in the console that reviews your workloads against best practices across the 6 pillars and provides improvement recommendations.

9 Exam Day Strategy

1. **Eliminate wrong answers first.** Usually 2 options are obviously wrong; choose between the remaining 2.
2. **Watch for qualifiers:** “MOST cost-effective”, “LEAST operational overhead”, “MOST secure” — each changes the answer.
3. **“No servers to manage”** ⇒ Serverless (Lambda, Fargate, Aurora Serverless, DynamoDB).
4. **“High availability”** ⇒ Multi-AZ, Auto Scaling, multiple regions.
5. **“Decouple”** ⇒ SQS (queue) or SNS (pub/sub).
6. **“Compliance reports”** ⇒ AWS Artifact (not GuardDuty, not Config).
7. **“Detect threats”** ⇒ GuardDuty; **“block web attacks”** ⇒ WAF; **“DDoS”** ⇒ Shield.
8. **“Audit API calls”** ⇒ CloudTrail; **“monitor metrics”** ⇒ CloudWatch; **“config changes”** ⇒ Config.
9. **“Migrate database”** ⇒ DMS; **“convert schema”** ⇒ SCT.
10. **Flag and skip** difficult questions; return at the end.
11. You have **90 minutes for 65 questions** — about 83 seconds per question. Keep moving.
12. **Mark for review:** If unsure, go with your gut then mark for review rather than spending too long.

Good luck on your exam!

AWS Certified Cloud Practitioner — CLF-C02